

PARTICULAR SPECIFICATION P2B

APPENDIX P2B – Interfaces with Other contractors

CONTENTS

P2B.1	Interface Work	1
P2B.1.1	General.....	1
P2B.1.2	Interfaces with Other contractors	2
P2B.1.3	Interfaces with C4A.....	2
P2B.1.4	Interfaces with IWMF	3
P2B.1.5	Interfaces to be Managed by the Contractor.....	4
P2B.2	Timing of Interface Works	5
P2B.2.1	Key Objectives	5
P2B.3	Requirements for Work in common areas and at interface Points	6
P2B.3.1	General.....	6
P2B.3.2	Concurrent Work at Interface Points.....	7
P2B.3.3	Sequential Work at Interface Points.....	8
P2B.4	Protection of Completed Infrastructure.....	8
P2B.5	Services Diversions and Protection	9
P2B.6	Interface with Receiving Contractors	9
P2B.7	Interface with Supplying Contractors.....	9
P2B.8	Interface Tie-in Schedules	10

NOTE: THIS DOCUMENT SUPERCEDES THE ORIGINAL VERSION OF THE APPENDIX P2B – INTERFACES WITH OTHER CONTRACTORS AT THE CLOSE OF TENDER. THE CHANGES HAS BEEN HIGHLIGHTED IN YELLOW.

P2B.1 INTERFACE WORK

P2B.1.1 General

- (a) The successful implementation of the TWRP Project is dependent on multiple contractors working together to build a complete and functional plant. There are inherent challenges associated with the proximity of works under different contractors, varying ground levels and timely completion of work for interfaces and integration with the balance of plant work. It is crucial for the successful and timely implementation of the TWRP Project that all the TWRP contracts are carried out to well-defined programmes that roll up to the overall TWRP programme meeting the milestone dates for completion. The S.O. will review programmes of work compiled by each contractor and will advise of any changes needed to ensure that the overall project programme objectives are achieved. On receipt of the S.O.'s review comments the Contractor shall make necessary revisions to his proposed programme to optimise coordination.
- (b) This section of the Specifications covers the Contractor's responsibilities to work with the S.O. to manage interfaces between the Works and the work of others. It also highlights design and construction considerations which are to be fully accounted for in the Contractor's proposed sequence of works, design and methods of construction.
- (c) The Board requires the Contractor to commit sufficient suitably-qualified and experienced resources to complete the project on time. Success in this respect will require the simultaneous carrying out of many parallel construction activities and this will require careful and detailed planning.
- (d) The Contractor shall, through liaison with the Board and the S.O., co-operate with other contractors, agencies and any other authorities.
- (e) The Contractor shall work in close coordination and collaboration with other contractors in the execution of the Works and especially for all interfaces and tie-in works.
- (f) The Contractor shall keep the Board and the S.O. fully informed on all matters involving co-operation and co-ordination with other contractors, agencies and any other authorities. All costs incurred in such co-operation and coordination, including adjusting the construction sequence, schedule, method statement and carrying out connections during off peak construction periods, night times or weekends to facilitate interface construction and connection are deemed to be included in the rates and prices.

P2B.1.2 Interfaces with Other contractors

- (a) The interfaces and tie-ins refer to all direct and indirect interfaces between the Works under this Contract and other contracts, agencies and any authorities. These include all the following contractors and their sub-contractors:
 - (i) All TWRP contractors (mainly C4A, C5A and C1A);
 - (ii) The appointed IMC contractor;
 - (iii) All contractors appointed by the Board;
 - (iv) All NEA contractors for the delivery of the IWMF;
 - (v) All authorities, agencies and service providers; and
 - (vi) Accredited Checkers, Registered Inspectors, Service Providers, etc.

P2B.1.3 Interfaces with C4A

- (a) The Contractor shall interface with Contract 4A, the Biosolids and Digestion contractor for the majority of the tie-ins and interfaces, except steam supply. As described in other sections of the specifications, the Digestion Pre-treatment is an integral part of the Biosolids and Digestion system as illustrated in Figure P2B.1 and the upstream and downstream of the Digestion Pre-treatment facility fully interacts with C4A's process units.
- (b) The following equipment will be located in, or routed through, the Biosolids Building and C4A controlled work areas. The Contractor shall work with the C4A contractor to plan and coordinate these works:
 - (i) THP feed pumps and DWAS pipework;
 - (ii) Compressed air system (For instrument air supply); and
 - (iii) Steam pipelines from the IWMF.

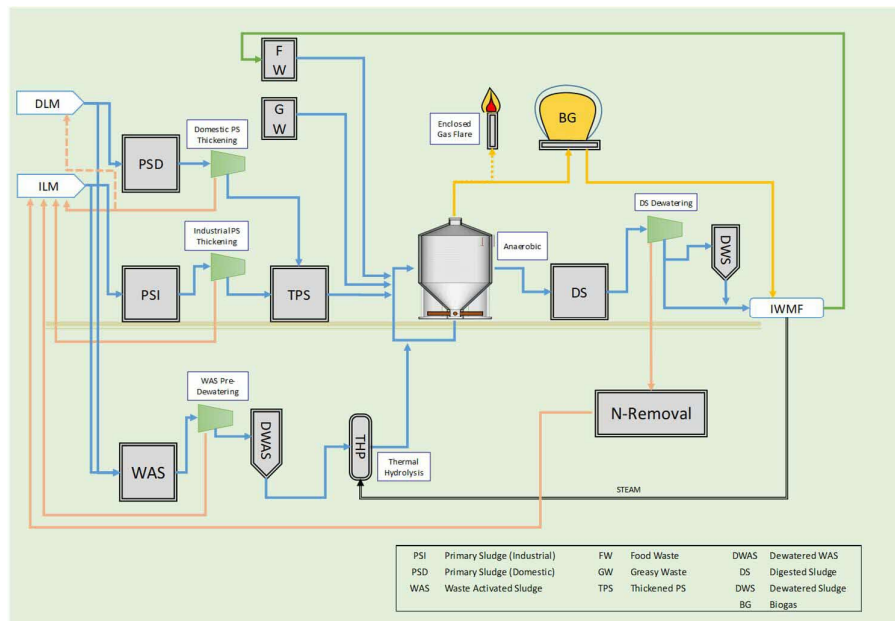


Figure P2B.1 Simplified Biosolids and Digesters Schematic

- (c) Included within this Appendix is a tie-in list of interfaces between C4B and C4A.

P2B.1.4 Interfaces with IWMF

- (a) The successful implementation of the TWRP Project and the associated synergies with the IWMF are dependent on contractors working together to build and integrate the two facilities.
- (b) The Contractor shall be responsible for the steam (STHP) process interface with IWMF. The steam (STHP) service tie-in is located on the DWS Conveyor and Pipe Bridge (by C4A) which connects the TWRP Biosolids Building and IWMF. The Contractor shall route the steam (STHP) services through the Biosolids Building to the THP courtyard area, thus the Contractor shall work with the C4A contractor to plan and coordinate the interface works.
- (c) DWS Conveyor and Pipe Bridge is a two-level structure with the dewatered sludge conveyors located on an incline deck and the pipes on a lower horizontal deck. The pipes located this bridge are:
- (i) 2 nos. Food Waste (FW) pipes; (by Other contractors)
 - (ii) 1 no. Condensate (SC) pipe; (by Other contractors)
 - (iii) 2 nos. Membrane Filtrate (MF) pipes; (by Other contractors) and
 - (iv) 2 nos. Steam (STHP) pipes.

- (d) The process interfaces shall be designed to be fail-safe: that is a sudden stop in consumption, or production, will not result in hazardous conditions. This means that the Contractor shall provide all the safety systems required without reliance on systems within IWMF.
- (e) The Contractor shall, through liaison with the Board and the S.O., co-operate with the C4A contractor, NEA contractors, agencies and any other authorities.
- (f) The Contractor shall work in close coordination and collaboration with the NEA contractors and C4A contractor in the execution of the Works and especially for all interfaces and tie-in works.
- (g) Included within this Appendix is a tie-in list of interfaces between C4B and IWMF. Data sheets providing details for each of these interfaces are provided.

P2B.1.5 Interfaces to be Managed by the Contractor

- (a) This section provides a brief overview of the interfaces and tie-ins with Other contractors under this Contract. The Contractor shall incorporate the requirements and interfaces as described in the other sections of the Specifications.
- (b) All pipe diameters shall not be changed for the entire length of the pipe routes, unless approved by the S.O..
- (c) The Contractor shall manage all interfaces for the Permanent Works under the Contract including but not limited to:
 - (i) Storm drains – Interfaces of drains with Other contractors;
 - (ii) Major yard piping – upstream and downstream interfaces of yard piping as shown in the model with Other contractors;
 - (iii) Process piping – interfaces as show in the Drawings with Other contractors;
 - (iv) Potable water mains – interfaces at potable water valve chambers and sub-meter chamber with Other contractors;
 - (v) Temporary NEWater water mains – interfaces at NEWater valve chambers and sub-meter chamber with Other contractors;
 - (vi) Fire water mains – interfaces at fire water ring main valve chambers with Other contractors;
 - (vii) Plant water – interfaces at plant water ring main valve chambers and sub-meter chamber with Other contractors;
 - (viii) Electrical distribution network cable duct banks and manholes – interfaces at cable duct banks with Other contractors; (Although there are no physical tie-ins of duct banks, the Contractor shall manage, coordinate and interface with Other contractors for electrical cabling inside the Biosolids Building);

- (ix) Electrical power supply interfaces between Contractor and Other contractors;
- (x) Instrumentation and controls interfaces between Contractor and Other contractors notably C5A contractor;
- (xi) IWMF Process interfaces - Steam (STHP); and
- (xii) Communication distribution network cable duct banks and manholes – interfaces at cable duct banks with Other contractors; (Although there are no physical tie-ins, the Contractor shall coordinate and interface with Other contractors for ICA cabling inside the Biosolids Building).
- (xiii) Building services equipment interfaces between Contractor and Other contractors;
- (xiv) Equipment installation – interfaces with Other Contractors;
- (xv) Structural interfaces between Contractor and Other contractors notably C4A contractor;
- (xvi) Openings, supports and hatches for equipment interfaces between Contractor and Other contractors; and
- (xvii) Cast-in items (if required) supply interfaces between Contractor and Other contractors.

P2B.2 TIMING OF INTERFACE WORKS

P2B.2.1 Key Objectives

- (a) The Contractor's sequence of work shall achieve the following key objectives in a logical sequence and on time to help achieve the overall project delivery programme, managed by the S.O.:
 - (i) Provide temporary and permanent connections to essential services – power, potable water, NEWater, sanitary and communications as early as possible to facilitate the Works as defined in the Specifications;
 - (ii) Complete the Contractor's temporary and permanent site drainage as early as possible to minimise any flooding risk in the TWRP Area;
 - (iii) Complete the Works to achieve the milestone(s) as defined in the Contract to facilitate the overall Project delivery. This includes providing design input and interfaces requirements as early as possible to facilitate the Works with Other Contractors; and
 - (iv) Complete the Testing and commissioning of the Works as defined in the Tender Documents to facilitate the overall Project delivery.

- (b) The Contractor shall work closely with the S.O., Other contractors and the NEA contractors to plan, coordinate and complete the interface works without impacts to the programme of works for the Contractor, Other contractors and the NEA contractors. This may include but not necessarily be limited to:
 - (i) Sharing of detailed programmes and sequences of work in advance for each tie-in interface;
 - (ii) Sharing of BIM models for the tie-in works;
 - (iii) Sharing of method statements, safety procedures and other relevant information concerning detailed construction methodologies to be employed for the tie-in;
 - (iv) Attending coordination meetings for the tie-in works;
 - (v) Joint inspections of the tie-in works prior to and after completion;
 - (vi) To timely advise of any information and requirements needed for the tie-in works and to provide information and requirements requested by the NEA contractors;
 - (vii) To provide sufficient access and working space for Other contractors when required for the tie-in works;
 - (viii) To re-instate the working area of Other contractors if required due to the tie-in works by the Contractor.

P2B.3 REQUIREMENTS FOR WORK IN COMMON AREAS AND AT INTERFACE POINTS

P2B.3.1 General

- (a) The Contractor shall be responsible for the Works up to the interface and tie – in point, including any protective works to prevent any damage to the completed Works.
- (b) The Contractor shall note that the Works in the Common Areas may have adjacent contract boundaries with Other contractors. The Contractor shall consider the schedule and works of the adjacent contractor(s) in the construction and installation of the Works in the Common Areas.
- (c) The Contractor shall ensure that all work is kept within the Contractor's work area and shall not encroach into the work area of Other contractors. Any works proposed to encroach into the work area of Other contractors shall be submitted to the S.O. for approval and agreed by Other contractors.
- (d) The Contractor shall be responsible for assessing the zone of influence of his works and shall design his work to minimise any influence on the work area of Other contractors subject to S.O. approval.

- (e) The Contractor shall provide schedules for all such interface works to the S.O. for approval.
- (f) The Contractor shall arrange with the S.O. for the inspection of the completed work prior to installing the protection to the interface point. The Contractor shall carry out any backfilling works after the inspections of the Works are completed.
- (g) For all buried pipework, the Contractor shall carry out the necessary pre-commissioning tests prior to backfilling.
- (h) For process pipework on pipe racks, the interface point requirements are as specified in P04 Mechanical Works and schedules to this Appendix.
- (i) All pipe interfaces shall be terminated at the interface point as specified in the tie-in lists. Where a bare end is specified, the first party to install the pipe up to the interface point shall leave a green end of at least 300mm for the interface connection.
- (j) Pipes shall be tested up to the interface point by the contractor installing the pipe. Only when the tests have been passed shall approval be given to make the final connection. Test documentation shall be provided to the contractor making the final connection.
- (k) The connection of pipe interfaces will be performed by the contractor whose works include the source of the fluid, as the party detailed in the “Contract From” field in the tie-in list.
- (l) The final testing of the pipe will be undertaken by the by the contractor whose works include the source of the fluid. In the event of a failure, a joint inspection by both contractors and the S.O. shall determine the responsibility for any rectification works. In the event of disagreement the S.O. shall advise the final decision. The Contractor shall allow for conducting two (2) retests.

P2B.3.2 Concurrent Work at Interface Points

- (a) At times, it will be necessary for the Contractor and Other contractors to implement work at the interface points at the same time to minimise any rework.
- (b) Upon review of the programmes by the various contractors, the S.O. will assess the feasibility of concurrent work at the interface points between the two or more contractors. When concurrent working is deemed feasible by the S.O., the Contractor shall work with the interfacing contractor(s) to coordinate the interface works.
- (c) The Contractor shall complete his work including all testing and inspections prior to completion of any interface works with the interfacing contractor(s).
- (d) The Contractor shall allow the interfacing contractor(s) a time window (subject to the S.O.’s approval) to work in his work area to complete such interfacing work.

P2B.3.3 Sequential Work at Interface Points

- (a) At times, it will be necessary for the Contractor to complete interface work, to which Other contractors will need to connect later.
- (b) The Contractor shall work with the S.O. to arrange inspection, testing and commissioning of the completed work prior to installing protection for the interface point.
- (c) The Contractor shall furnish as-built details to the S.O. for approval for his interfacing works.
- (d) The Other contractors will carry out the ERSS and excavation works to expose the interface point and its protection elements and arrange for an inspection with the S.O. and the Contractor prior to any works to the interface point.
- (e) The Other contractors will allow the Contractor an agreed time window (subject to approval by the S.O.) to work in his work area to remove the protection elements of the interface. The Contractor shall arrange for an inspection with the S.O. and the Other contractor once the protection elements of the interface have been removed.
- (f) The Others contractor will proceed with the completion of the interfacing works after the inspections are cleared.
- (g) The Other contractors will backfill the excavation at the interfacing area only after the inspection, testing and commissioning of the interfacing point.

P2B.4 PROTECTION OF COMPLETED INFRASTRUCTURE

- (a) The Contractor shall note that all completed infrastructure under the Contract shall remain the responsibility of the Contractor from the time of completion to the substantial completion of the whole of the Works. The Contractor shall provide all necessary protection measures to prevent any damage to the completed works whether by the Contractor or by Other contractors.
- (b) The Contractor shall note that there will be infrastructure completed by Other contractors in the Contractors working area and shared working areas and the Contractor shall exercise due diligence so as not to damage these infrastructures. Any damage by the Contractor to the Other contractor's infrastructure shall be repaired and reinstated to the original condition, to the approval of the S.O. The Contractor shall provide the same warranty to the repaired infrastructure (as applicable) to the approval of the S.O..
- (c) The Contractor shall be responsible for making good any damages in the completed Works between the Works completion and substantial completion of the Works.

P2B.5 SERVICES DIVERSIONS AND PROTECTION

- (a) The Contractor shall make all necessary services diversions where these may be affected by or affect the progress of the Works.
- (b) The Contractor shall conduct services detection for the proposed development and definition of all pipe and cable routes. All existing services shall be clearly pegged at Site for the S.O.'s inspection. The Contractor shall submit any proposed services diversion and protection plans to the S.O. for approval prior to the commencement of any site work. Trial holes shall be carried out to ascertain the position, depths and sizes of the services where necessary.

P2B.6 INTERFACE WITH RECEIVING CONTRACTORS

- (a) Where applicable, the Contractor (Supplying Contractor) shall work with the S.O. to co-ordinate and interface with the Other contractor (Receiving Contractor) to provide the specified equipment (free issue) to the Receiving Contractor for incorporation in the Works. The Contractor shall make timely arrangements for meetings to plan the delivery of the equipment and any inspection(s) required, with the Receiving Contractor and the S.O..
- (b) The Contractor shall store the equipment in accordance with the manufacturer's instructions and shall be visibly marked or identified in a manner as directed by the S.O. and shall be properly protected from loss and damage until receipt of the equipment by the Receiving Contractor.
- (c) The Receiving Contractor shall transport the specified equipment from the Contractors storage location (to be agreed by the S.O.) to the required facility for installation by the Receiving Contractor.
- (d) The Contractor shall refer to the additional interface requirements in Particular Specifications.

P2B.7 INTERFACE WITH SUPPLYING CONTRACTORS

- (a) The Contractor shall work with the S.O. to co-ordinate and interface with the Supplying Contractor to receive the specified equipment (free issue) to the Contractor for incorporation in the Works. The Contractor shall make timely arrangements for meetings to plan the receipt of the equipment and any inspection(s) required, with the Supplying Contractors and the S.O..
- (b) The Contractor shall store the equipment in accordance with the manufacturer's instructions and shall be visibly marked or identified in a manner as directed by the S.O. and shall be properly protected from loss and damage.
- (c) The Contractor shall transport the specified equipment from the agreed Supplying Contractors storage location to the required facility for installation by the Contractor. The Contractor shall allow for all costs associated with the transport, handling, and installation of the specified free -issued equipment.

- (d) The Contractor shall refer to the Particular Specification P6 for the hardware items to be supplied by the Supplying Contractor (C5A contractor).
- (e) The Contractor shall not move the received equipment from Site without the written approval from the S.O..

P2B.8 INTERFACE TIE-IN SCHEDULES

- (a) C4B Yard Piping Interfaces;
- (b) C4B G2D Plant Water Interfaces;
- (c) C4B PW FPW Interfaces;
- (d) C4B Surface Water Interfaces;
- (e) C4B ICA Interfaces; and
- (f) C4B Electrical Interfaces.

END OF SECTION

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND SIDESTREAM NITROGEN REMOVAL FACILITIES (2ND-STAGE)

VOLUME 3

C4B Yard Piping Interfaces

Service	Service Code	Pipe Description	Pipe Diameter	Pipe Material	Full Tag Number	Contract From	Contract To	Location	Tie-in Connection	Max Instantaneous Flow (l/s)	Interface Pressure (kPag)	PID
Foul Air	FA	Foul air from THP to digesters	80	SS16	WSD11-TP-FA-0101	C4B	C4A	THP courtyard	Bare pipe - 300mm green end	300 Nm3/h	320	WSD02-P-6100
		Foul air from THP to digesters	80	SS16	WSD11-TP-FA-0102	C4B	C4A	THP courtyard	Bare pipe - 300mm green end	300 Nm3/h	320	WSD02-P-6100
		Foul air from THP Cake Feed Pumps Inlet Hoppers to FA main header	80	FRP	WSD11-TP-FA-0201	C4B	C4A	THP Feed Pump room	Flanged	By Contractor	By Contractor	WSD11-P-6000
		Foul air from Future THP Cake Feed Pump Inlet Hoppers to FA main header	By Contractor	FRP	WSD11-TP-FA-0402	FUT	C4B	THP courtyard	Flanged	By Contractor	By Contractor	WSD11-P-6000
		Foul air from Future THP Cake Feed Pump Inlet Hoppers to FA main header	By Contractor	FRP	WSD11-TP-FA-0401	FUT	C4B	THP courtyard	Flanged	By Contractor	By Contractor	WSD11-P-6000
		Foul air from Future THP Cake Feed Pump Inlet Hoppers to FA main header	By Contractor	FRP	WSD11-TP-FA-0302	FUT	C4B	THP courtyard	Flanged	By Contractor	By Contractor	WSD11-P-6000
		Foul air from Future THP Cake Feed Pump Inlet Hoppers to FA main header	By Contractor	FRP	WSD11-TP-FA-0301	FUT	C4B	THP courtyard	Flanged	By Contractor	By Contractor	WSD11-P-6000
Industrial Water	IW	ILM 1 IW Tertiary Plant output to IW Storage	600	FRP	WIL13-TP-IW-0102	C3B1	C4B	Link Bridge between Side stream N to ILM	Flanged	596	278	WIL13-P-6080
		ILM 1 IW Tertiary Plant output to IW Storage	600	FRP	WIL13-TP-IW-0103	C4B	C4A	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	596	250	WIL13-P-6080
		ILM 2 IW Tertiary Plant output to IW Storage	600	FRP	WIL23-TP-IW-0202	C3B1	C4B	Link Bridge between Side stream N to ILM	Flanged	596	278	WIL23-P-6080
		ILM 2 IW Tertiary Plant output to IW Storage	600	FRP	WIL23-TP-IW-0203	C4B	C4A	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	596	250	WIL23-P-6080
Plant Drain	PD	Plant Drain and Recycles to ILM Headworks	350	SS16	WSB01-TP-PD-0103	C4A	C4B	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	155	60	WSB01-P-6090
		Plant Drain and Recycles to ILM Headworks	350	SS16	WST02-TP-PD-0101	C4B	C3B1	Link Bridge between Side stream N to ILM	Flanged	155	60	WIL11-P-6000
		Sidestream N Plant Drainage to TWRP Sewer	300	DIPU	WST00-TP-PD-0101	C4B	C1A	TWRP Plant drain manhole PMDH-0102	Socket	171	atm	WSW00-P-6001
		Plant Drain pipe from THP modules to PD header	150	SS16	WSD11-TP-PD-0102	C4B	C4A	Apron Slab adjacent to courtyard	Stub	By Contractor	atm	WSD11-P-6010
		From Future THP 3 to plant drain header	By Contractor	SS16	WSD11-TP-PD-0401	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6040	
		From Future THP 3 to plant drain header	By Contractor	SS16	WSD11-TP-PD-0301	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6030	
Plant Water (DLM MF)	G2D	Plant water 1 to THP	300	SS16	WSB01-TP-G2D-0101	C4A	C4B	THP courtyard	Bare pipe - 300mm green end	128	532	WSB01-P-6200
		Plant water 2 to THP	300	SS16	WSB01-TP-G2D-0102	C4A	C4B	THP courtyard	Bare pipe - 300mm green end	128	532	WSB01-P-6200
		Plant Water to Sidestream N Treatment Below Ground	300	DIPU	WST00-TP-G2D-0101	C1A	C4B	East of WST00	Socket	20	528	WSW00-P-6200
		From THP G2D ringmain to THP 4 (Future)	By Contractor	SS16	WSD11-TP-G2D-0401	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6040	
		From THP G2D ringmain to Future THP 3 HS Stage 2 HEX 1	By Contractor	SS16	WSD11-TP-G2D-0304	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6030	
		From THP G2D ringmain to THP 3 (Future)	By Contractor	SS16	WSD11-TP-G2D-0301	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6040	
		From THP G2D ringmain to Future THP 3 HS Stage 1 HEX 1	By Contractor	SS16	WSD11-TP-G2D-0302	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6030	
		From THP G2D ringmain to Future THP 3 HS Stage 1 HEX 2	By Contractor	SS16	WSD11-TP-G2D-0303	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6030	
		From THP G2D ringmain to Future THP 3 HS Stage 2 HEX 2	By Contractor	SS16	WSD11-TP-G2D-0305	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6030	
		From THP G2D ringmain to Future THP 4 HS HEX	By Contractor	SS16	WSD11-TP-G2D-0401	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6040	
		From THP G2D ringmain to Future THP 4 HS HEX	By Contractor	SS16	WSD11-TP-G2D-0402	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6040	
		From THP G2D ringmain to Future THP - Spare	By Contractor	SS16	WSD11-TP-G2D-0101	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6200	
		From THP G2D ringmain to Future THP - Spare	By Contractor	SS16	WSD11-TP-G2D-0201	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6200	
Primary Sludge Industrial	PSI	ILM1 to PSI Holding Tank	250	SS16	WSB01-TP-PSI-0101	C4B	C4A	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	46	726.2	WSB01-P-6010
		ILM2 to PSI Holding Tank	250	SS16	WSB01-TP-PSI-0201	C4B	C4A	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	46	726.2	WSB01-P-6010
		ILM2 Primary sludge to PSI Holding Tank	250	SS16	WIL22-TP-PSI-0109	C3B1	C4B	Link Bridge between Side stream N to ILM	Flanged	46	726.2	WIL22-P-6023
		ILM1 Primary Sludge to PSI Holding Tank	250	SS16	WIL12-TP-PSI-0101	C3B1	C4B	Link Bridge between Side stream N to ILM	Flanged	46	726.2	WIL12-P-6023
Steam High Pressure	STHP	STHP to THP 1	150	SS16	WSB01-TP-STHP-0101	IWMF	C4B	TWRP and IWMF boundary - 1st Tier of Conveyor Bridge	Bare pipe - 300mm green end	9.1 tonne/h	1220	WSD11-P-6050
		STHP to THP 2	150	SS16	WSB01-TP-STHP-0201	IWMF	C4B	TWRP and IWMF boundary - 1st Tier of Conveyor Bridge	Bare pipe - 300mm green end	9.1 tonne/h	1220	WSD11-P-6050
		STHP to Future THP Phase 2	150	SS16	WSD11-TP-STHP-0102	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	9.1 tonne/h	1220	WSD11-P-6050
		STHP to Future THP Phase 2	150	SS16	WSD11-TP-STHP-0202	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	9.1 tonne/h	1220	WSD11-P-6050
		STHP to Future THP 3 (Phase 1)	By Contractor	SS16	WSD11-TP-STHP-0301	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	1220	WSD11-P-6050
		STHP to Future THP 4 (Phase 1)	By Contractor	SS16	WSD11-TP-STHP-0401	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	1220	WSD11-P-6050
Thickening Centrate Industrial	TCI	TCI Pumping to ILM Grit Tank Effluent Channel	750	CSEL	WSB01-TP-TCI-0102	C4A	C4B	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	323	50	WSB01-P-6081
		Thickening Centrate to ILM1 Grit Tank Effluent Channel	500	CSEL	WSB01-TP-TCI-0101	C4B	C3B1	Link Bridge between Side stream N to ILM	Flanged	502	50	WIL11-P-6082
Waste Activated Sludge Industrial	WASI	ILM1 to WAS Holding Tank	250	SS16	WSB01-TP-WASI-0101	C4B	C4A	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	146	692	WSB01-P-6020
		ILM2 to WAS Holding Tank	250	SS16	WSB01-TP-WASI-0201	C4B	C4A	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	146	692	WSB01-P-6020
		ILM2 WAS header to Biosolids treatment facility	250	SS16	WIL23-TP-WASI-0204	C3B1	C4B	Link Bridge between Side stream N to ILM	Flanged	146	692	WIL23-P-6042
		ILM1 WAS header to Biosolids treatment facility	250	SS16	WIL13-TP-WASI-0204	C3B1	C4B	Link Bridge between Side stream N to ILM	Flanged	146	692	WIL13-P-6042
RO Permeate	ROP	ROP to THP for Steam Desuperheater	50	SS16	WSD11-TP-ROP-0101	C4A	C4B	North West of Biosolids Building L1	Flanged with blind flanged	1	230 to 300	WSD11-P-6050
		ROP to THP for Steam Desuperheater	50	SS16	WSD11-TP-ROP-0102	C3D2	C4B	Link bridge between Biosolids building and DLM2	Flanged with blind flanged	1	230 to 300	WSD11-P-6050
Dewatering Centrate	DC	Dewatering centrate to ILM1 Grit Tank Effluent Channel	400	CSEL	WST02-TP-DC-0101	C4B	C3B1	Link Bridge between Side stream N to ILM	Flanged	160	150	WIL11-P-6008
		DC to Sidestream N	250	CSEL	WSB01-TP-DC-0101	C4A	C4B	Link Bridge between Biosolids to Side stream N	Bare pipe - 300mm green end	80	230	WSB01-P-6082
		Dewatering centrate to Future Discharge Transfer pump	By Contractor	CSEL	WST02-TP-DC-0201	C4B	FUT	SSN Pump room	Flanged	By Contractor	By Contractor	WST02-P-6003
		Dewatering centrate from Future Discharge Transfer pump	By Contractor	CSEL	WST02-TP-DC-0202	C4B	FUT	SSN Pump room	Flanged	By Contractor	By Contractor	WST02-P-6003
Dewatered Waste Activated Sludge	DWAS	DWAS Silo 1 Discharge Chute		SS16	WSB01-TP-DWAS-0101	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6026
		DWAS Silo 1 Discharge Chute		SS16	WSB01-TP-DWAS-0102	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6026
		DWAS Silo 1 Discharge Chute		SS16	WSB01-TP-DWAS-0103	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6026
		DWAS Silo 1 Discharge Chute		SS16	WSB01-TP-DWAS-0104	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6026
		DWAS Silo 2 Discharge Chute		SS16	WSB01-TP-DWAS-0201	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6027
		DWAS Silo 2 Discharge Chute		SS16	WSB01-TP-DWAS-0202	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6027
		DWAS Silo 2 Discharge Chute		SS16	WSB01-TP-DWAS-0203	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6027
		DWAS Silo 2 Discharge Chute		SS16	WSB01-TP-DWAS-0204	C4A	C4B	Biosolids Building L1, Beneath DWAS silo	Flanged with slide gate	6	60	WSB01-P-6027
		DWAS Silo 1 Discharge Chute to Future THP feed pump 5	By Contractor	SS16	WSD11-TP-DWAS-0301	C4B	FUT	Biosolids Building L1, Beneath DWAS silo	Flanged	6	60	WSD11-P-6000
		DWAS Silo 1 Discharge Chute to Future THP feed pump 6	By Contractor	SS16	WSD11-TP-DWAS-0302	C4B	FUT	Biosolids Building L1, Beneath DWAS silo	Flanged	6	60	WSD11-P-6000
		DWAS Silo 1 Discharge Chute to Future THP feed pump 7	By Contractor	SS16	WSD11-TP-DWAS-0401	C4B	FUT	Biosolids Building L1, Beneath DWAS silo	Flanged	6	60	WSD11-P-6000
DWAS Silo 1 Discharge Chute to Future THP feed pump 8	By Contractor	SS16	WSD11-TP-DWAS-0402	C4B	FUT	Biosolids Building L1, Beneath DWAS silo	Flanged	6	60	WSD11-P-6000		
Instrument Air	IA	Instrument air to Biosolids building 1	80	SS16	WSD11-TP-IA-0101	C4B	C4A	Biosolids Building L1	Bare pipe - 300mm green end	TBC	TBC	WSB01-P-6600
		Instrument air to Biosolids building 2	80	SS16	WSD11-TP-IA-0102	C4B	C4A	Biosolids Building L1	Bare pipe - 300mm green end	TBC	TBC	WSB01-P-6600
		From THP IA header to THP 4 (Future)	By Contractor	SS16	WSD11-TP-IA-0401	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6040	
		From THP IA header to THP 3 (Future)	By Contractor	SS16	WSD11-TP-IA-0301	C4B	FUT	THP courtyard	By Contractor	By Contractor	WSD11-P-6030	
Hot Water (Return)	HWR	Hot water return from Heat Exchangers to TCD Wet Well	250	SS16	WSD11-TP-HWR-0101	C4B	C4A	THP courtyard	Bare pipe - 300mm green end	119	150	WSB01-P-6080
		Hot water return from Heat Exchangers to GW transfer pipe	150	SS16	WSD11-TP-HWR-0201	C4B	C4A	THP courtyard	Bare pipe - 300mm green end	28	350	WSD05-P-6021

TUAS WATER RECLAMATION PLANT
 CONTRACT 4B – DIGESTION PRE-TREATMENT AND SIDESTREAM NITROGEN REMOVAL FACILITIES (2ND-STAGE)

VOLUME 3

C4B Yard Piping Interfaces

Service	Service Code	Pipe Description	Pipe Diameter	Pipe Material	Full Tag Number	Contract From	Contract To	Location	Tie-in Connection	Max Instantaneous Flow (l/s)	Interface Pressure (kPag)	PID
Hot Water (Return)	HWR	Hot Water to THP 4 HS Stage 2 HEX 2	By Contractor	SS16	WSD11-TP-HWR-0404	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
		Hot Water to THP 3 HS Stage 2 HEX 2	By Contractor	SS16	WSD11-TP-HWR-0304	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
		Hot Water to THP 4 HS Stage 1 HEX 1	By Contractor	SS16	WSD11-TP-HWR-0401	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
		Hot Water to THP 4 HS Stage 2 HEX 1	By Contractor	SS16	WSD11-TP-HWR-0403	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
		Hot Water to THP 4 HS Stage 1 HEX 2	By Contractor	SS16	WSD11-TP-HWR-0402	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
		Hot Water to THP 3 HS Stage 1 HEX 1	By Contractor	SS16	WSD11-TP-HWR-0301	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
		Hot Water to THP 3 HS Stage 2 HEX 1	By Contractor	SS16	WSD11-TP-HWR-0303	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
		Hot Water to THP 3 HS Stage 1 HEX 2	By Contractor	SS16	WSD11-TP-HWR-0302	C4B	FUT	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSB01-P-6080
Hydrolysed Sludge	HS											
		Hydrolysed sludge feed to digesters	150	SS16	WSD11-TP-HS-0101	C4B	C4A	THP courtyard	Bare pipe - 300mm green end	17	400	WSD02-P-6100
		Hydrolysed sludge feed to digesters	150	SS16	WSD11-TP-HS-0102	C4B	C4A	THP courtyard	Bare pipe - 300mm green end	17	400	WSD02-P-6100
		Hydrolysed sludge from Future THP 3 Digester feed pumps	By Contractor	SS16	WSD11-TP-HS-0301	FUT	C4B	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSD02-P-6100
		Hydrolysed sludge from Future THP 4 Digester feed pumps	By Contractor	SS16	WSD11-TP-HS-0401	FUT	C4B	THP courtyard	Bare pipe - 300mm green end	By Contractor	By Contractor	WSD02-P-6100

C4B G2D Plant Water Interfaces

Service	Service Code	Pipe Description	Pipe Diameter	Pipe Material	Full Tag Number	Contract From	Contract To	Location	Tie-in Connection	Max Instantaneous Flow (l/s)	Interface Pressure (kPag)	PID
Plant Water (DLM MF)	G2D											
		Plant water 1 to THP	300	SS16	WSB01-TP-G2D-0101	C4A	C4B	THP courtyard	Bare pipe - 300mm green end	128	532	WSB01-P-6200
		Plant water 2 to THP	300	SS16	WSB01-TP-G2D-0102	C4A	C4B	THP courtyard	Bare pipe - 300mm green end	127.62	532	WSB01-P-6200
		Plant Water to Sidestream N Treatment Below Ground	300	DIPU	WST00-TP-G2D-0101	C1A	C4B	East of WST00	Socket	20	528	WSW00-P-6200
		From THP G2D ringmain to THP 4 (Future)	By Contractor	SS16	WSD11-TP-G2D-0401	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6040
		From THP G2D ringmain to Future THP 3 HS Stage 2 HEX 1	By Contractor	SS16	WSD11-TP-G2D-0304	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6030
		From THP G2D ringmain to THP 3 (Future)	By Contractor	SS16	WSD11-TP-G2D-0301	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6040
		From THP G2D ringmain to Future THP 3 HS Stage 1 HEX 1	By Contractor	SS16	WSD11-TP-G2D-0302	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6030
		From THP G2D ringmain to Future THP 3 HS Stage 1 HEX 2	By Contractor	SS16	WSD11-TP-G2D-0303	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6030
		From THP G2D ringmain to Future THP 3 HS Stage 2 HEX 2	By Contractor	SS16	WSD11-TP-G2D-0305	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6030
		From THP G2D ringmain to Future THP 4 HS HEX	By Contractor	SS16	WSD11-TP-G2D-0401	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6040
					WSD11-TP-G2D-0402	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6040
		From THP G2D ringmain to Future THP - Spare	By Contractor	SS16	WSD11-TP-G2D-0101	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6200
					WSD11-TP-G2D-0201	C4B	FUT	THP courtyard	By Contractor	By Contractor	By Contractor	WSD11-P-6200

C4B PW FPW Interfaces

Service	Service Code	Description	Pipe Diameter	Pipe Material	Full Tag Number	Contract From	Contract To	Location	Tie-in Connection
Potable Water	PW	Potable Water	200	DIPU	WST02-TP-PW-0101	C1A	C4B	South/East of WST02	Socket connection + Plug
			50	SS16	WSD11-TP-PW-0101	C1A	C4B	South /East of THP Courtyard	Flanged connection
Fire Hydrant Water	FH	Fire Water	200	DICL	WSW41-TP-FH-0101	C1A	C4B	South /East of WST02 Sidestream N Building	Flanged connection + Blank Flange

C4B Surface Water Interfaces

Service	Service Code	Description	Full Tag Number	Contract From	Contract To	Location
Surface Water Drain	WD	minor drain	WSW04-TP-WD-0101	C4B	C1A	South of Future WST02
		minor drain	WSW04-TP-WD-0103	C4B	C1A	South of Future WST02

C4B ICA Interfaces

Project Prefix	Facility	Contract From	Contract To	Location	Service	Typical	Description	Full Tag Number	Tie-in Connection	ScopeComb
WST02	Sidestream Nitrogen Removal	C1A	C4B	East of Sidestream Nitrogen Removal	ICA	ICA F/O	JDUCT	WST02-TP-J-1001	Refer to Volume 6 for Duct Bank Typical Details	C1AC4B
				East of Sidestream Nitrogen Removal		ICA F/O	JDUCT	WST02-TP-J-1002	Refer to Volume 6 for Duct Bank Typical Details	C1AC4B

C4B Electrical Interfaces

Project Prefix	Facility	Contract From	Contract To	Location	Service	Description	Full Tag Number	Tie-in Connection
WST02	Sidestream Nitrogen Removal	C3A	C4B	N side of the WST02	Electricity	EDUCT	WST02-TP-E-0001	Refer to Volume 6 for Duct Bank Typical Details

END OF SECTION